ISM Companion Reader Session 8

Central Limit Theorem & Confidence Intervals

Sampling distributions, standard error, point estimation, and interval bounds.

1 Sampling Distributions & The Central Limit Theorem

In real-world data analysis, we rarely observe an entire population. Instead, we select a random sample and compute a summary statistic, such as the sample mean $\bar{X}$. Because different random samples produce slightly different sample means, the statistic $\bar{X}$ itself behaves as a random variable with its own probability distribution—known as the **sampling distribution of the mean**.

Why Sampling Distributions Matter

Imagine measuring customer ages across multiple stores. A single sample gives one average age. If you repeat this process 1,000 times, those 1,000 averages form a neat curve. The Central Limit Theorem guarantees that as long as your sample size n is moderately large ($n \geq 30$), this curve becomes normal, no matter how skewed the original population was.

1.1 The Central Limit Theorem (CLT)

For any population with mean μ and standard deviation σ, the sampling distribution of the sample mean $\bar{X}$ approaches a Normal distribution with:

$$E[\bar{X}] = \mu$$

$$\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}} \quad \text{(Standard Error)}$$

The standardized Z-score for the sample mean is given by:

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$$

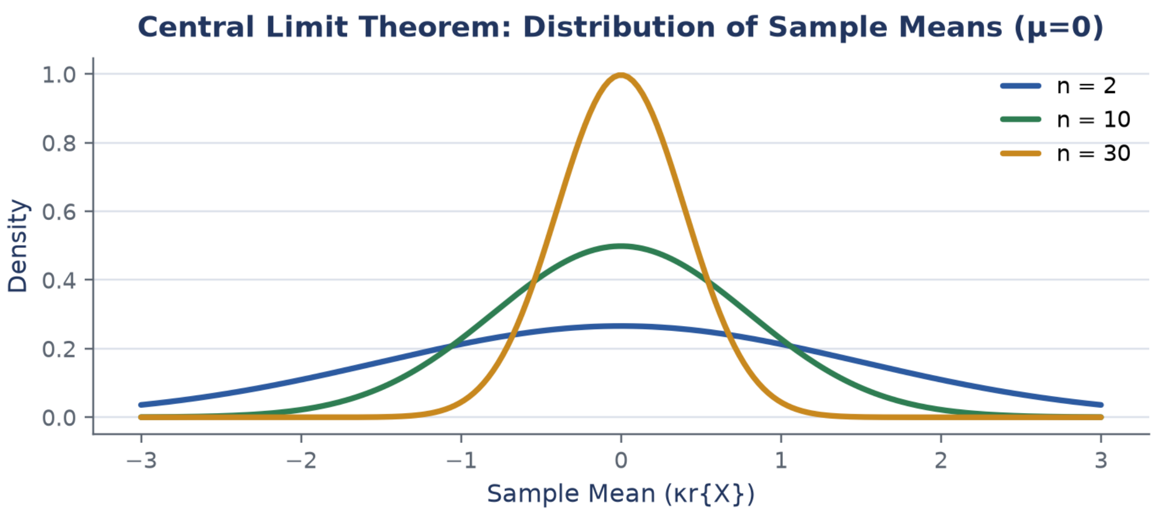

Figure: As sample size n increases, the distribution of sample means concentrates tightly around the population mean μ.

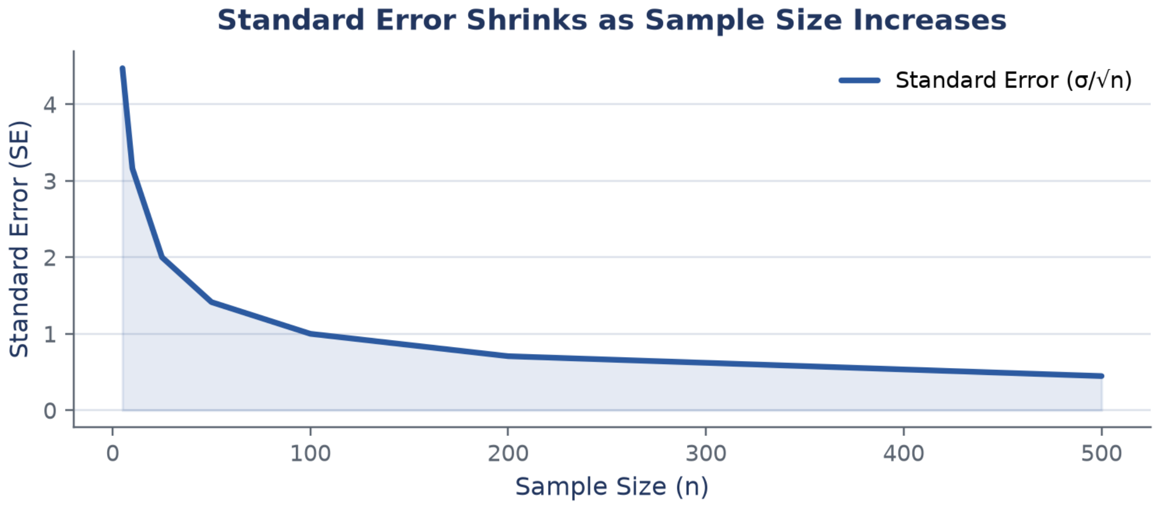

Figure: Standard Error drops proportionally to $1/\sqrt{n}$, showing dimishing returns for larger sample sizes.

Slide Example 1: Vehicle Registration Age

Problem Statement: The average age of registered vehicles in the US is 8 years (96 months) with a standard deviation of 16 months. A random sample of $n = 64$ vehicles is drawn. What is the probability that the sample mean age is between 90 and 100 months?

Step 1: Calculate the Standard Error:

$$\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}} = \frac{16}{\sqrt{64}} = \frac{16}{8} = 2 \text{ months}$$

Step 2: Convert bounds to Z-scores:

$$Z_1 = \frac{90 - 96}{2} = -3.00, \quad Z_2 = \frac{100 - 96}{2} = 2.00$$

Step 3: Find table areas: $P(Z \leq 2.00) = 0.9772$ and $P(Z \leq -3.00) = 0.0013$.

Step 4: Compute interval probability:

$$P(90 \leq \bar{X} \leq 100) = 0.9772 - 0.0013 = 0.9759 \quad (97.59\%)$$

2 Sampling Distribution of Sample Proportions

When measuring categorical outcomes (e.g., proportion of satisfied customers), the sample proportion $\bar{p} = X/n$ follows a normal distribution for large samples ($np \geq 5$ and $n(1-p) \geq 5$).

Mean and Standard Error for Proportions:

$$E[\bar{p}] = p, \quad \sigma_{\bar{p}} = \sqrt{\frac{p(1-p)}{n}}$$

3 Point Estimation & Confidence Intervals

A **point estimate** provides a single best guess for an unknown parameter (e.g., sample mean $\bar{X}$ estimates μ). However, point estimates rarely hit the exact parameter. A **confidence interval** adds a margin of error around the point estimate, offering a range of plausible values at a specified confidence level $(1 - \alpha)$.

Confidence Interval Formula for Population Mean μ (σ known):

$$\text{CI} = \bar{X} \pm Z_{\alpha/2} \left(\frac{\sigma}{\sqrt{n}} \right)$$

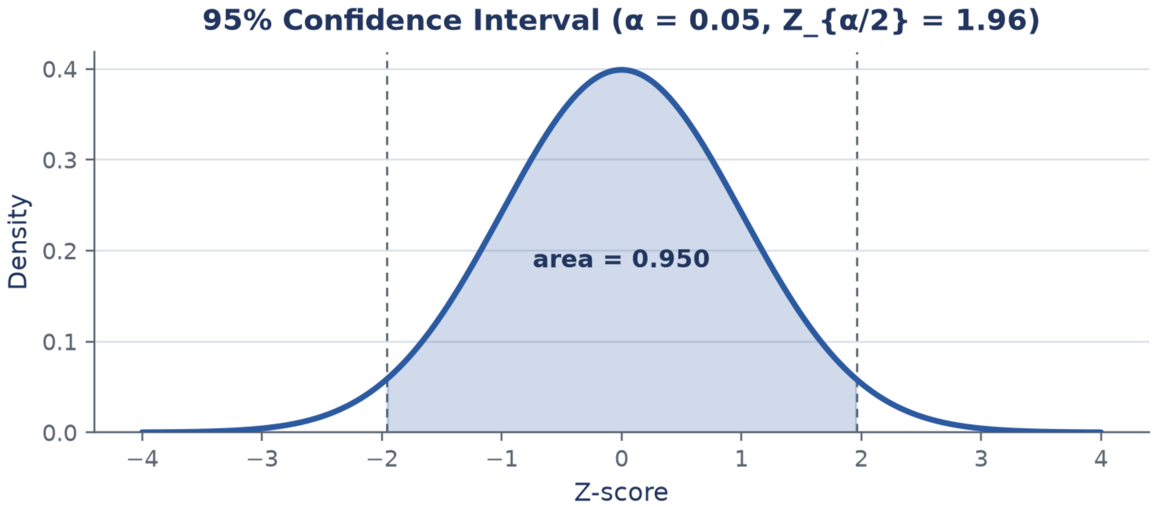

Figure: A 95% confidence interval leaves $\alpha/2 = 0.025$ in each tail, corresponding to critical $Z_{\alpha/2} = 1.96$.

Slide Example: Tire Store Expenditure

Problem Statement: Mean expenditure at a tire store is $\bar{X} = \$85.00$ with population standard deviation $\sigma = \$12.00$ based on a sample of $n = 36$ customers. Construct a 95% confidence interval for the true mean expenditure.

Step 1: Identify parameters and critical value: $\bar{X} = 85$, $\sigma = 12$, $n = 36$. For 95% confidence level ($\alpha = 0.05$), $Z_{\alpha/2} = 1.96$.

Step 2: Calculate Standard Error:

$$\sigma_{\bar{X}} = \frac{12}{\sqrt{36}} = \frac{12}{6} = 2.00$$

Step 3: Compute Margin of Error:

$$\text{ME} = 1.96 \times 2.00 = 3.92$$

Step 4: Formulate interval bounds:

$$\text{Lower Bound} = 85.00 - 3.92 = 81.08, \quad \text{Upper Bound} = 85.00 + 3.92 = 88.92$$

Thus, we are 95% confident that the population mean expenditure is between \$81.08 and \$88.92.

Interpreting Confidence Intervals

Never say "there is a 95% probability that the population mean lies in this specific interval." The true mean is a fixed number, not a random variable. The 95% probability refers to the long-run procedure: 95% of intervals generated across many samples will contain μ.

Summary Checklist

- **Central Limit Theorem:** For $n \geq 30$, sample means $\bar{X}$ follow $N(\mu, \sigma/\sqrt{n})$ regardless of population shape.
- **Standard Error:** $\sigma_{\bar{X}} = \sigma/\sqrt{n}$ measures the variation of sample means around μ.
- **Confidence Interval:** $\bar{X} \pm Z_{\alpha/2}(\sigma/\sqrt{n})$ balances point estimate with precision.